

{20-09-25}

Date _____
Page _____

Tutorial 48 → Naive Bayes' Classifier

Dataset

$$X = \{x_1, x_2, \dots, x_n\} \{y\}$$

Bayes' Theorem

$$P(A/B) = \frac{P(B/A) \times P(A)}{P(B)}$$

$$P\left(\frac{y}{x_1, x_2, \dots, x_n}\right) = \frac{P(x_1/y) P(x_2/y) \dots P(x_n/y) * P(y)}{P(x_1) P(x_2) \dots P(x_n)}$$

$$\left\{ \frac{P(y) \cdot \prod_{i=1}^n P(x_i/y)}{P(x_1) P(x_2) \dots P(x_n)} \right\}$$

↳ Const. same for every records.

$$\therefore P\left(\frac{y}{x_1, x_2, \dots, x_n}\right) \propto P(y) \cdot \prod_{i=1}^n P(x_i/y)$$

$$y = \underset{\text{argmax}}{y} P(y) \prod_{i=1}^n P(x_i/y)$$

↳ Considering max of TEMPERATURE

OUTLOOK					TEMPERATURE				
	Yes	No	P(Y)	P(N)		Yes	No	P(Y)	P(N)
Sunny	2	3	2/9	3/5	Hot	2	2	2/9	2/5
Overcast	4	0	4/9	0/5	Mild	4	2	4/9	2/5
Rainy	3	2	3/9	2/5	Cold	3	1	3/9	1/5
Total	9	5	100%	100%	Total	9	5	100	100.

PLAY

		P(Y) & P(N)
Yes	9	9/14
No	5	5/14
Total	14	100%

Today (Sunny, Hot)

Date _____
Page _____

$$P(\text{Yes} / \text{Today}) = \frac{P(\text{Today Sunny} / Y) \times P(\text{Hot} / Y) \times P(Y)}{P(\text{Today})} \rightarrow \text{cancel}$$

$$\Rightarrow \frac{2}{9} \times \frac{2}{9} \times \frac{1}{147}$$

$$\Rightarrow \frac{2}{63} \rightarrow 0.031$$

$$P(\text{No} / \text{Today}) = \frac{P(\text{Sunny} / N) \times P(\text{Hot} / N) \times P(N)}{P(\text{Today})}$$

$$\Rightarrow \frac{3}{5} \times \frac{2}{9} \times \frac{9}{147}$$

$$\Rightarrow 0.08571$$

$$\text{Now, } P(Y_{\text{no}}) = \frac{0.031}{0.031 + 0.08571}$$

$$\approx 0.27$$

$$P(N_{\text{no}}) = 1 - 0.27$$

$$\Rightarrow 0.73$$

$$\hookrightarrow P(N_{\text{no}}) > P(Y_{\text{no}})$$

So, No Play Today!